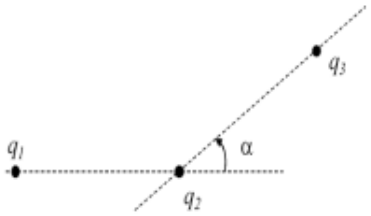
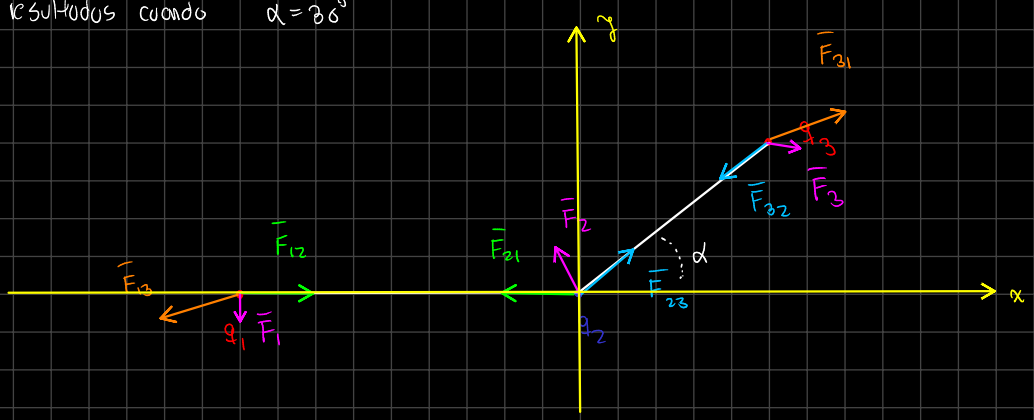
2. ¿Qué fuerza se debe ejercer sobre cada una de las tres cargas puntuales de la figura (q_1 =1 μC, q_2 = -2 μC y q_3 = 0,5 μC) para que el sistema esté en equilibrio?

La distancia entre q_1 y q_2 es igual a la distancia entre q_3 y q_2 y vale 30 cm. Analizar los resultados cuando el ángulo α = 30°.



Datos: $q_1 = 1\mu C$ $q_2 = -2\mu C$ $q_3 = 0.5\mu C$ $d = 30cm$ $\alpha = 30°$

Analizar los resultados cuando $\alpha = 30°$



$$\bar{F}_1 + \bar{F}_2 + \bar{F}_3 = 0$$

$$\bar{F}_{12} + \bar{F}_{13} + \bar{F}_{21} + \bar{F}_{23} + \bar{F}_{31} + \bar{F}_{32} = 0$$

Solo debo calcular $\bar{F}_{12}$, $\bar{F}_{13}$ y $\bar{F}_{23}$

$$\bar{F}_{12} = k\, q_1\, \frac{q_2\,(\bar{r}_1 - \bar{r}_2)}{|\bar{r}_1 - \bar{r}_2|^3}$$

$\bar{r}_1 = -d\hat{\imath}$ $\quad \bar{r}_1 - \bar{r}_2 = -d\hat{\imath}$

$\bar{r}_2 = \bar{0}$ $\quad |\bar{r}_1 - \bar{r}_2| = d$

$$\bar{F}_{12} = k\, q_1\, \frac{q_2 \cdot -d\hat{\imath}}{d^3}$$

$$\bar{F}_{12} = -k\, \frac{q_1\, q_2}{d^2}\,\hat{\imath}$$

$$\bar{F}_{12} = -9\cdot10^9\ \frac{Nm^2}{C^2} \times \frac{(1\mu C)(-2\mu C)}{(0.3m)^2}\,\hat{\imath}$$

$$\bar{F}_{12} = -9\cdot10^9\ \frac{Nm^2}{C^2} \times \frac{(1)(-2)\,\mu^2 C^2}{(0.3)^3\,m^2}\,\hat{\imath}$$

$$\bar{F}_{12} = -9\cdot10^9\ N \times \frac{(-2)(10^{-6})^2}{(0.3)^2}\,\hat{\imath}$$

$$\bar{F}_{12} = 0.2\ N\,\hat{\imath}$$

$$\bar{F}_{23} = K\, q_2\, \frac{q_3\,(\bar{r}_2 - \bar{r}_3)}{|\bar{r}_2 - \bar{r}_3|^3}$$

$\bar{r}_2 = 0$

$\bar{r}_3 = d\cos(\alpha)\,\hat{i} + d\,\mathrm{sen}(\alpha)\,\hat{j}$

$\bar{r}_2 - \bar{r}_3 = -d\cos(\alpha)\,\hat{i} - d\,\mathrm{sen}(\alpha)\,\hat{j}$

$|\bar{r}_2 - \bar{r}_3| = d$

$$\bar{F}_{23} = K\, q_2\, \frac{q_3\,(-d\cos(\alpha)\,\hat{i} - d\,\mathrm{sen}(\alpha)\,\hat{j})}{d^3}$$

$$\bar{F}_{23} = K\, q_2\, q_3\, \frac{(-\cos(\alpha)\,\hat{i} - \mathrm{sen}(\alpha)\,\hat{j})}{d^2}$$

$$\bar{F}_{23} = 9\cdot 10^9\ \frac{Nm^2}{C^2} \times \frac{(-2\mu C)(0.5\mu C)\,(-\cos(30°)\,\hat{i} - \mathrm{sen}(30°)\,\hat{j})}{(0.3\,m)^2}$$

$$\boxed{\bar{F}_{23} = 0.0866\,N\,\hat{i} + 0.05\,N\,\hat{j}}$$

Puedo calcular $\bar{F}_2 = \bar{F}_{21} + \bar{F}_{23} = -\bar{F}_{12} + \bar{F}_{23}$

$= -0.2\,N\,\hat{i} + 0.0866\,N\,\hat{i} + 0.05\,N\,\hat{j}$

$$\boxed{\bar{F}_2 = -0.1134\,N\,\hat{i} + 0.05\,N\,\hat{j}}$$

Ahora hay que calcular $\bar{F}_{13}$

$$\bar{F}_{13} = \frac{K\, q_1\, q_3\,(\bar{r}_1 - \bar{r}_3)}{|\bar{r}_1 - \bar{r}_3|^3}$$

$\#\ \bar{r}_1 = -d\,\hat{i}$ $\quad\#\ \bar{r}_3 = d\cos(\alpha)\,\hat{i} + d\,\mathrm{sen}(\alpha)\,\hat{j}$

$\#\ \bar{r}_1 - \bar{r}_3 = -d\,\hat{i} - d\cos(\alpha)\,\hat{i} - d\,\mathrm{sen}(\alpha)\,\hat{j} = (-d - d\cos(\alpha))\,\hat{i} - d\,\mathrm{sen}(\alpha)\,\hat{j}$

$= -d(1+\cos(\alpha))\,\hat{i} - d\,\mathrm{sen}(\alpha)\,\hat{j}$

$\#\ |\bar{r}_1 - \bar{r}_3|^3 = \sqrt{d^2(1+\cos(\alpha))^2 + d^2\,\mathrm{sen}^2(\alpha)}$

$= \sqrt{d^2(1 + 2\cos(\alpha) + \cos^2(\alpha)) + d^2\,\mathrm{sen}^2(\alpha)}$

$= \sqrt{d^2 + 2d^2\cos(\alpha) + d^2\cos^2(\alpha) + d^2\,\mathrm{sen}^2(\alpha)}$

$= \sqrt{2d^2 + 2d^2\cos(\alpha)}$

$= \sqrt{2d^2(1+\cos(\alpha))}^3$

$q_1 = 1\mu C \qquad q_2 = -2\mu C \qquad q_3 = 0.5\mu C \qquad d = 30\,cm \qquad \alpha = 30°$

$$\bar{F}_{13} = \frac{k\, q_1\, q_3 \left[-d(1+\cos(\alpha))\,\hat{i} - d\,\mathrm{Sen}(\alpha)\,\hat{j}\right]}{\sqrt{2d^2(1+\cos(\alpha))}^{\,3}}$$

$$F_{13x} = \frac{k\, q_1\, q_3 \left[-d(1+\cos(\alpha))\,\hat{i}\right]}{\left[2d^2(1+\cos(\alpha))\right]^{3/2}}$$

$$F_{13x} = 9\cdot 10^9 \frac{Nm^2}{C^2} \times \frac{(1\mu C)(0.5\mu C)\left[-0.3m\,(1+\cos(30°))\,\hat{i}\right]}{\left[2(0.3m)^2(1+\cos(30°))\right]^{3/2}} \rightarrow 0.19466375$$

$$F_{13x} = -0.01294\,N\,\hat{i}$$

$$F_{13y} = \frac{k\, q_1\, q_3 \left[-d\,\mathrm{Sen}(\alpha)\,\hat{j}\right]}{\left[2d^2(1+\cos(\alpha))\right]^{3/2}}$$

$$F_{13y} = 9\cdot 10^9 \frac{Nm^2}{C^2} \times \frac{(1\mu C)(0.5\mu C)\left[-0.3m\,\mathrm{Sen}(30°)\,\hat{j}\right]}{\left[2(0.3m)^2(1+\cos(30°))\right]^{3/2}} \rightarrow 0.19466375$$

$$F_{13y} = -0.0034675\,N\,\hat{j}$$

$$\boxed{\bar{F}_{13} = -0.01294\,N\,\hat{i} - 0.0034675\,\hat{j}}$$

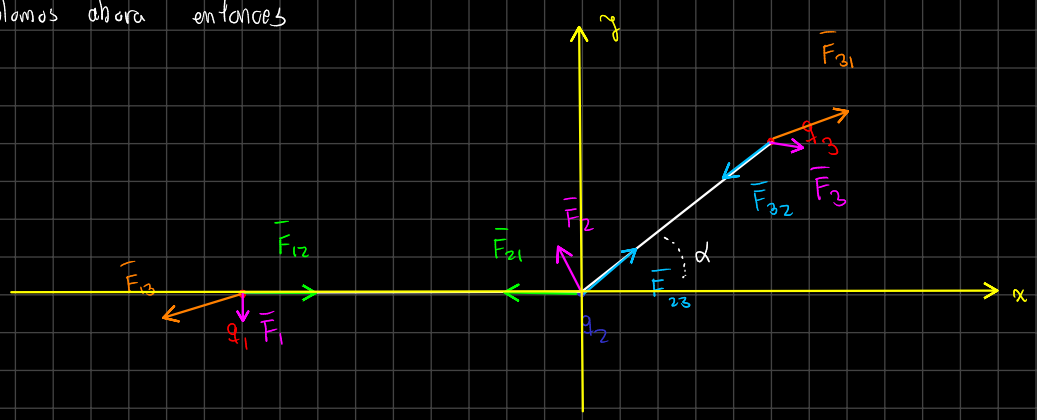
Podre Calcular $\bar{F}_1$

$$\bar{F}_1 = F_{12} + F_{13}$$

$$= 0.2N\,\hat{i} - 0.01294\,N\,\hat{i} - 0.0034675\,\hat{j}$$

$$\boxed{\bar{F}_1 = 0.18706\,N\,\hat{i} - 0.0034675\,\hat{j}}$$

Calculamos ahora entonces



$$\bar{F}_3 = \bar{F}_{31} + \bar{F}_{32} = -\bar{F}_{13} - \bar{F}_{23}$$

$$= 0.01294\,N\hat{i} + 0.0034675\,\hat{j} - 0.0866\,N\hat{i} - 0.05N\hat{j}$$

$$= -0.07366\,N\hat{i} - 0.0465N\hat{j}$$

$$\boxed{\bar{F}_3 = -0.0736N\hat{i} - 0.0465N\hat{j}}$$

Ô Verificando los resultados

$$\boxed{\bar{F}_1 = 0.18706\,N\hat{i} - 0.0034675\hat{j}}$$

$$\boxed{\bar{F}_2 = -0.1134\,N\hat{i} + 0.05N\hat{j}}$$

$$\boxed{\bar{F}_3 = -0.0736N\hat{i} - 0.0465N\hat{j}}$$